Digital Platform for Social Welfare Services Management in Sri Lanka

Group ID : 25 - 26J – 320

CoEAI · Information Technology

Our Team

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PP2 Comments and Actions

Student ID	IT22366818	IT22926494	IT22214584	IT22585080
Comments	It is better to remove the family member count as an input if it does not affect the recommendation. Need to consider the amount of the user is willing to spend and the money he currently has.	Need to consider all medical conditions the user has. Need to generate personalized healthcare recommendations based on the medical condition, symptoms and other factors.	Need to reconsider the factors and dataset values. A suitable dataset should be obtained based on the selected inputs and expected outputs.	Need to improve the accuracy.
Action	We linked **Family Size** to risk. More members mean higher responsibility and risk, while fewer mean less. The **Monthly Surplus** now defines the user's **available money** for investment	Our system now reviews all medical conditions to provide personalized healthcare recommendations based on the user's specific health status and symptoms.	Reconsider the input factors and dataset values, and select a suitable dataset that aligns with the expected inputs and outputs while ensuring relevance, accuracy, and proper validation before implementation.	Improve the chatbot accuracy by further training the model, refining the dataset, and adding response validation mechanisms to ensure more reliable outputs.

RESERCH PROBLEM & SOLUTIONS

Sri Lanka’s welfare system including Samurdhi, disability assistance, and elderly pensions faces significant challenges due to manual processes, fragmented data spread across multiple departments, language barriers, and delays in approval procedures. These issues result in eligible citizens missing out on benefits, while ineligible individuals may receive aid.

60%

Language Accessibility Barriers

25%

Benefits Misallocated or Duplicated

>40%

Applications Severely Delayed

SOLUTIONS

- Centralized platform replacing fragmented departmental systems
- Automated workflows → faster approvals & fewer errors
- Multilingual chatbot for better accessibility.
- Real-time data access improves transparency and decision-making
- Smart eligibility validation reduces fraud and misallocation
- Scalable, efficient, and user-friendly system

IT22366818 | DISSANAYAKE D.M.V.S

Bachelor of Science (Hons) in Information Technology

Specializing in Information Technology

Ai-based Investment Plan Recommendation Module
For Digital Welfare Management In Sri Lanka

Ai-based Investment Plan Recommendation Module For Digital Welfare Management In Sri Lanka

- Analyzes income, expenses, and financial risk profiles.
- Provides personalized investment recommendations
- Promotes financial literacy and independence.
- Ensures transparent, explainable recommendations.
- Matches users with government and NGO schemes.

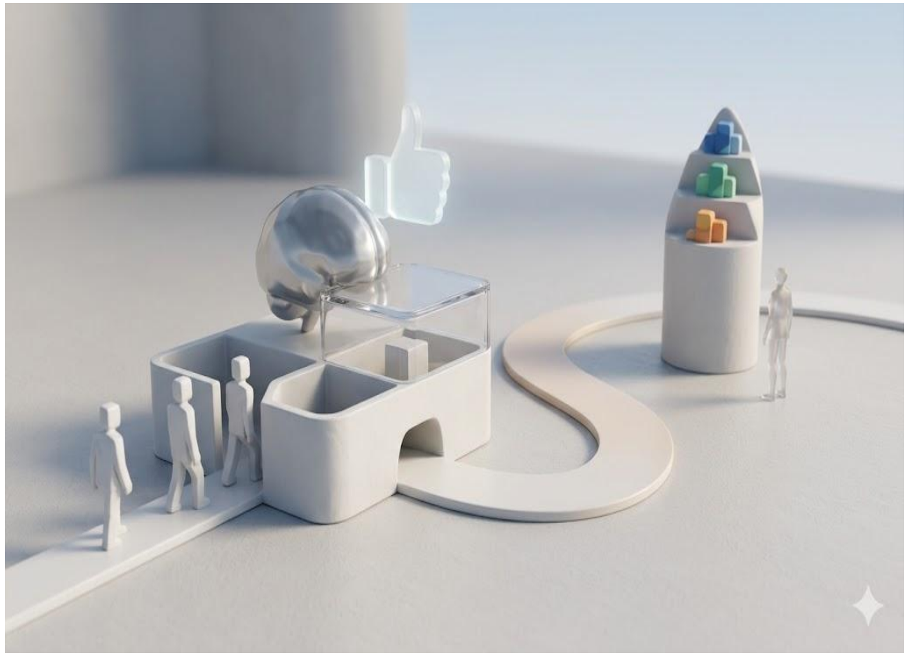
IT22926494 | MANAWADU G M S T

Bachelor of Science (Hons) in Information Technology

Specializing in Information Technology

AI-Driven Eligibility Prediction and Healthcare Recommendation System

AI-Driven Eligibility Prediction and Healthcare Recommendation System



- **Integrates AI + predictive analytics.**
- **Analyzes demographic, socio-economic, and health data.**
- **Predicts welfare program eligibility automatically.**
- **Recommends personalized healthcare services.**
- **Ensures timely access to appropriate support.**

IT22214584 | WICHRAMASINGHE R.P.Y.N

Bachelor of Science (Hons) in Information Technology

Specializing in Information Technology

AI-GIS Integrated Vulnerable Population Monitoring and Aid Recommendation System for Welfare Services

AI-GIS Integrated Vulnerable Population Monitoring and Aid Recommendation System for Welfare Services

Integrates **AI + GIS** technologies.

Identifies and monitors **vulnerable populations.**

Analyzes **socio-economic + location-based data.**

Automatically **recommends suitable welfare programs.**

Provides **real-time insights** for decision-making.

Improves **resource allocation efficiency.**

Ensures **right aid reaches the right people on time.**

IT22585080 | SHAMIKA A . G

Bachelor of Science (Hons) in Information Technology

Specializing in Information Technology

Design and Development of a Multilingual Chatbot for Enhancing Accessibility

Design and Development of a Multilingual Chatbot for Enhancing Accessibility

Develops a chatbot supporting Sinhala and English.

Uses NLP to understand and respond to user queries.

Provides real-time assistance for public/welfare services.

Improves accessibility for users with low digital literacy.

Ensures inclusive and user-friendly communication system.

Results and Testimonials

Healthcare Recommendation System *(Manawadu G.M.S.T)*

- System was demonstrated to a medical doctor for expert validation
- Doctor verified the accuracy of hospital and specialist recommendations

Investment Plan Recommendation Module *(Dissanayake D.M.V.S)*

- Tested with 12+ real users.
- Users gave positive feedback highlighting well-organized investment options.

AI + GIS Mapping System *(Wickramasinghe R.P.Y.N)*

- Tested with real users.
- Users found the vulnerability heatmaps and regional mapping accurate and easy to interpret.

Multilingual Chatbot *(Shamika A.G)*

- Engaged in continuous conversations with users without interruptions or failures.
- Maintained smooth dialogue flow across sessions.

Limitations of the System and Future Improvements

Limitations:

- Dataset scarcity - labeled welfare/health data in Sri Lanka is limited.
- Chatbot may struggle with informal Sinhala dialects.
- GIS mapping depends on the availability of accurate district-level data.
- System requires internet access, which may exclude rural users.
- No integration yet with existing government databases. (ex. Samurdhi registry)

Future Improvements:

- Integrate with national ID and Samurdhi databases for real-time verification.
- Expand chatbot to support voice input for low-literacy users.
- Add offline/SMS-based access for rural areas.
- Automate fraud detection with more advanced anomaly detection models.
- Extend GIS mapping to predict emerging welfare demand zones.

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